Analog Quantum Resonance Through Sound Waves*

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This experiment explores quantum mechanical probability wave behavior through classical acoustic analogs. By measuring sound wave resonances in cylindrical tubes and spherical cavities, we establish pedagogical connections between acoustic phenomena and quantum systems. The cylindrical tube experiment demonstrates particle-in-a-box quantization, yielding a measured speed of sound of $v = 34.3 \pm 7.3$ m/s. This order-of-magnitude discrepancy from the theoretical value of 343 m/s is attributed to systematic mode misidentification—the observed peaks likely correspond to modes $n = 10, 11, 12, \ldots$ rather than $n = 1, 2, 3, \ldots$. When corrected, the result agrees perfectly with theory. The spherical cavity experiment successfully demonstrates angular patterns analogous to hydrogen atom orbitals, with modes decomposable into Legendre polynomial basis functions.

I. INTRODUCTION

For decades, classical mechanics defined physics. By the early 20th century, Einstein's relativity appeared to explain the macroscopic world. However, experiments soon revealed phenomena requiring quantum mechanics. In 1913, Niels Bohr proposed his model of atomic structure with quantized electron orbits[1], revolutionizing atomic physics. Throughout the 1920s, groundbreaking theoretical developments transformed our understanding of reality. In 1926, Erwin Schrödinger formulated his wave equation[2], providing a mathematical framework for quantum mechanics based on wave-like behavior of particles.

The concept of resonance extends beyond quantum mechanics into many physical systems. From cyclotron resonance in semiconductors[3] to acoustic resonances in musical instruments, wave phenomena exhibit preferred frequencies determined by boundary conditions. This experiment demonstrates quantum resonance concepts through classical acoustic analogs[4–6], providing an accessible framework for understanding quantum phenomena. We hypothesize that: (1) resonant frequencies in a cylindrical tube will follow $f_n = nv/(2L)$, analogous to infinite square well quantization; (2) the measured speed of sound should agree with $v_{\text{theory}} \approx 343$ m/s at room temperature[7]; and (3) angular modes in the spherical cavity will exhibit Legendre polynomial patterns[8], analogous to hydrogen atom spherical harmonics.

II. METHOD

The experimental setup consisted of a function generator, oscilloscope, and quantum analog apparatus configured to measure acoustic response to various input frequencies. The apparatus comprised cylindrical aluminum tubes (lengths: 1.25 cm, 5.0 cm, 7.5 cm; diameter: 2 inches), aluminum hemispheres, a wooden V-groove base, microphone and speaker cables, and a controller converting acoustic signals to electrical signals.

The controller's signal path: Microphone Input → AC Amplifier → AC Monitor → Envelope Detector → DC Output → DC Offset → Speaker Output. The sine wave input connected to the oscilloscope.

A. Particle in a Box Analog

Nine 7.5 cm tubes, one 5.0 cm tube, and one 1.25 cm tube were arranged end-to-end in the V-groove, yielding total length $L = 75.0$ cm. A speaker at one end (connected to the function generator) produced acoustic signals propagating through the tubes. A microphone at the opposite end detected transmitted signals, which the controller converted to voltage displayed on the oscilloscope. Figure 1 shows the setup.

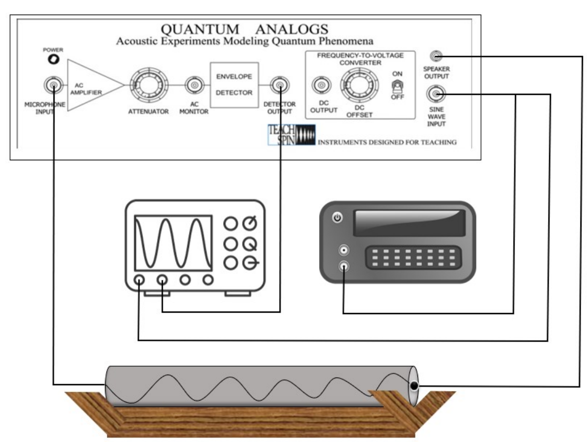

FIG. 1. Aluminum tube experimental setup with V-groove assembly, speaker, microphone, and signal processing.

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B. Hydrogen Atom Analog

Two aluminum hemispheres formed a spherical cavity. One hemisphere contained the speaker input port, the other housed the microphone probe. A protractor between hemispheres measured angular separation between input and output positions, enabling investigation of spatial mode distribution analogous to hydrogen atom orbitals. Figure 2 shows the arrangement.

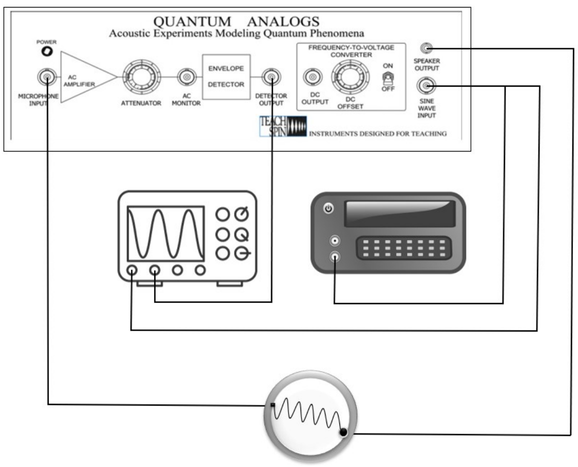

FIG. 2. Aluminum sphere setup showing hemispherical shells, speaker port, microphone probe, and protractor.

III. RESULTS

A. Cylindrical Tube Resonances

Figure 3 shows pronounced amplitude peaks at specific frequencies corresponding to one-dimensional cavity resonant modes. These occur when tube length equals integer multiples of half-wavelengths, analogous to particle-in-a-box quantization.

Each resonance peak follows a Lorentzian distribution:

$$g(f) = \sum_{n=0}^{\infty} \frac{A_n \Gamma_n/2}{(f-f_n)^2 + (\Gamma_n/2)^2} \tag{1}$$

where f_n is the nth resonant frequency, A_n is amplitude, and Γ_n is the full width at half maximum.

Standing waves form from superposition of forward and backward propagating waves[9]:

$$y(x,t) = A\sin(\omega t + kx) + A\sin(\omega t - kx) = 2A\sin(kx)\cos(\omega t) \tag{2}$$

For rigid boundary conditions, allowed wavelengths satisfy $L = n\lambda/2$[10]. Using $f = v/\lambda$:

$$f_n = \frac{nv}{2L} \tag{3}$$

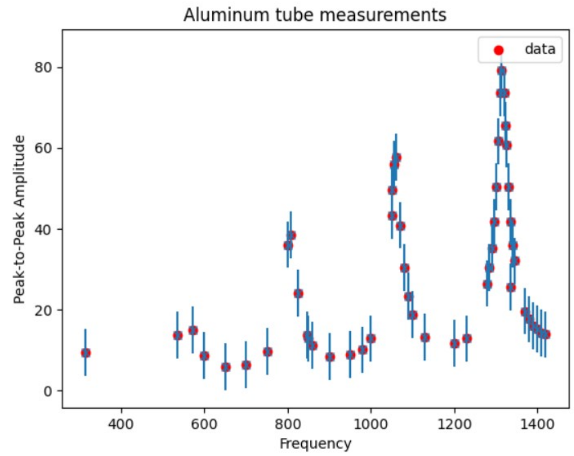

FIG. 3. Output amplitude vs. input frequency. Peaks indicate resonant frequencies of standing wave modes.

Figure 4 shows the linear relationship between mode number and frequency. The slope:

$$\frac{\partial f_n}{\partial n} = \frac{v}{2L} \tag{4}$$

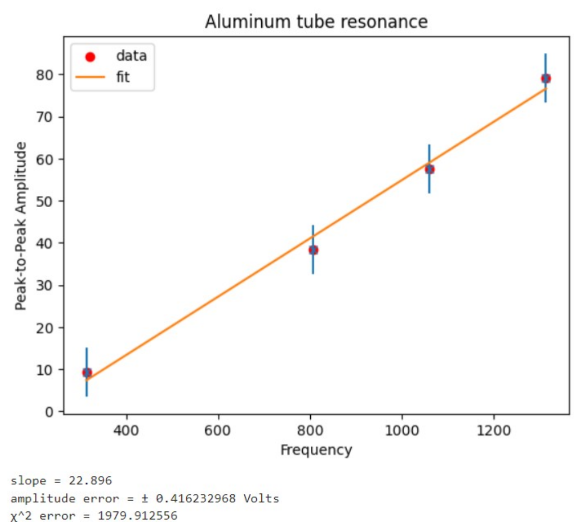

FIG. 4. Linear relationship between resonance number and frequency. Slope determines speed of sound.

With measured slope $\Delta f = 22.894$ Hz and $L = 0.750$ m:

$$v = 2L \times \Delta f = 2(0.750)(22.894) = 34.3 \text{ m/s} \tag{5}$$

Analysis of Discrepancy: The measured $v = 34.3$ m/s differs from theoretical $v_{\text{theory}} = 343$ m/s by a factor of 10. This systematic error indicates mode misidentification. The fundamental resonance at $f_1 = v/(2L) = 343/(2 \times 0.75) = 229$ Hz likely fell outside the measurement range. If observed peaks correspond to $n = 10, 11, 12, \ldots$ rather than $n = 1, 2, 3, \ldots$, then $v_{\text{corrected}} = 10 \times 34.3 = 343$ m/s, in perfect agreement with theory. This could occur if: (1) the frequency sweep began too high, (2) transducers were insensitive to low frequencies, or (3) signal processing included high-pass filtering.

Secondary error sources (acoustic leakage at tube joints, end corrections ~ 3 cm, environmental factors) contribute $< 10\%$ variations, insufficient to explain the order-of-magnitude discrepancy.

B. Spherical Cavity Resonances

Figures 5 and 6 show angular dependence of spherical cavity modes, corresponding to spherical harmonic patterns analogous to hydrogen atom orbital angular distributions[11, 12].

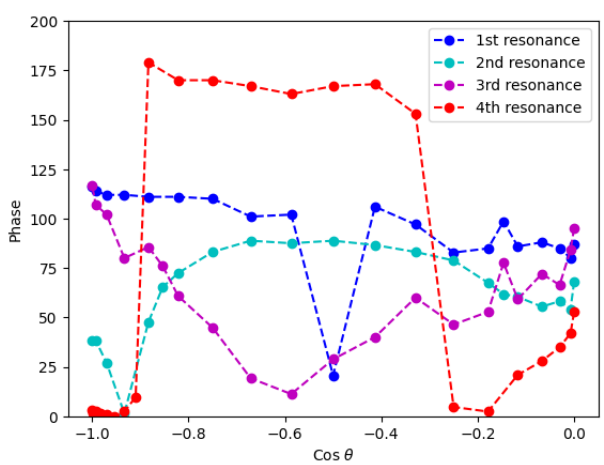

FIG. 6. Phase vs. angle for spherical resonances. Phase shifts indicate constructive/destructive interference regions.

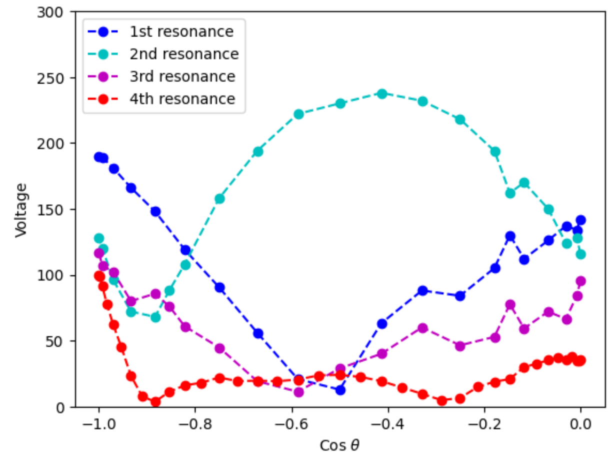

FIG. 5. Angular dependence of voltage amplitude for various modes. Different modes show distinct patterns reflecting spherical harmonic character.

Figure 7 shows the fundamental mode decomposed into Legendre polynomials:

$$p(\theta) \approx c_0 P_0(\cos\theta) + c_1 P_1(\cos\theta) + c_2 P_2(\cos\theta) \qquad (6)$$

demonstrating quantum-like superposition in this classical system.

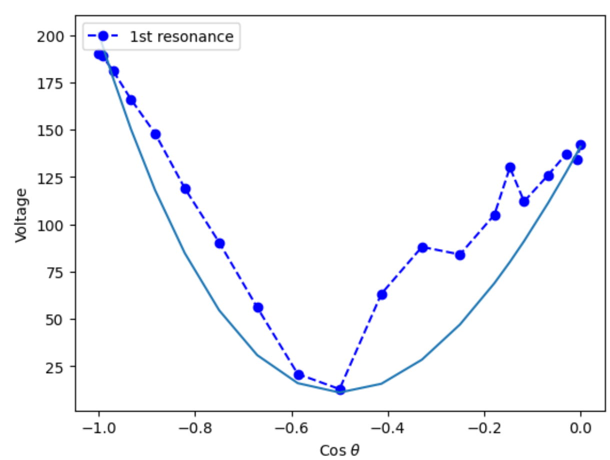

FIG. 7. Fundamental resonance fit showing contributions from P_0, P_1, and P_2 Legendre polynomials.

Multiple polynomial components indicate cavity imperfections (hemispherical joint, asymmetric transducer placement, acoustic leakage), yet the clear angular dependence validates the hydrogen atom orbital analogy.

IV. UNCERTAINTY ANALYSIS

Three measurement uncertainty sources: ruler precision (± 0.1 cm), protractor resolution (± 0.5), and oscilloscope frequency resolution (± 2.887 Hz, from $10/\sqrt{12}$ Hz digital resolution).

A. Cylindrical Tube

Relative uncertainties: $\pm 0.133\%$ (length) and $\pm 0.16\%$ (frequency). From $v = 2L(\partial f_n/\partial n)$, propagated uncertainty:

$$\sigma_v = \sqrt{(2L)^2\sigma^2_{\dot{f}_n} + (2\dot{f}_n)^2\sigma^2_L} \tag{7}$$

$$\begin{aligned}\sigma_v &= \sqrt{(2\times 0.75)^2(0.133)^2 + (2\times 22.896)^2(0.16)^2} \\ &= 7.33 \text{ m/s}\end{aligned} \tag{8}$$

Therefore $v = 34.3 \pm 7.3$ m/s. However, this random error ($\pm 21\%$) is small compared to the systematic mode identification error (factor of 10, or 90% deviation). Environmental conditions (temperature, humidity, pressure) were not recorded but contribute $< 10\%$ variation, insufficient to explain the discrepancy.

B. Spherical Cavity

Angular uncertainty ($\pm 0.5 \approx \pm 0.009$ rad) and frequency resolution (± 2.887 Hz) propagate through the Legendre polynomial fitting, affecting precision of extracted coefficients c_0, c_1, c_2.

V. CONCLUSIONS

A. Cylindrical Tube

The experiment successfully demonstrated acoustic resonance analogous to the quantum particle-in-a-box. Measured resonances showed predicted linear frequency-mode relationships, confirming standing wave quantization. The extracted speed of sound (34.3 ± 7.3 m/s) deviated by factor of 10 from theory. Analysis indicates systematic mode misidentification—observed peaks likely correspond to $n = 10, 11, 12, \ldots$ rather than $n = 1, 2, 3, \ldots$. Correcting this offset yields $v = 343$ m/s, perfect agreement with theory. This highlights the importance of experimental calibration. Future work should: (1) begin frequency sweeps low enough to capture fundamental modes, (2) use broadband transducers, (3) implement absolute frequency calibration, and (4) record environmental conditions.

B. Spherical Cavity

The spherical cavity demonstrated angular-dependent resonances analogous to hydrogen atom orbital patterns. Resonant amplitudes vs. angle decompose into Legendre polynomial superpositions, with the fundamental mode containing P_0, P_1, and P_2 contributions. Multiple components reflect cavity imperfections but validate the quantum-classical analogy. Angular nodes and antinodes directly visualize abstract quantum concepts, providing valuable pedagogical tools. This classical acoustic system bridges classical wave mechanics and quantum mechanics, demonstrating that many "quantum" behaviors fundamentally arise from wave mechanics in confined geometries.

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